



Faculty of Informatics & Design
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Assessment Two: Research Proposal Submission

TITLE OF RESEARCH PROPOSAL

The Adoption of Cloud Computing by Small to Medium Enterprises in South Africa

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1. Introduction to the Research

In recent years, technology has transformed the approaches and directions in which many businesses operate in this era of technological advancement. The widespread move towards cloud computing represents one of the most significant developments in this transformation. Cloud computing offers organisations the affordability to run software applications, manage business systems, and store and retrieve information enabled by internet connectivity, without the need to invest in expensive physical infrastructure (Jayeola et al., 2022:228; Mokwena & Sithole, 2024:12; Mkhize et al., 2025:5). According to the National Institute of Standards and Technology, cloud computing is defined as a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort (Mokwena & Sithole, 2024:12).

In practical terms, cloud computing is delivered through three principal service models. The first is Software as a Service (SaaS), which allows users to access applications such as financial reporting tools or communication and emailing systems directly over reliable internet connectivity without local installation. The second is Platform as a Service (PaaS), which provides developers with an online environment to build and deploy applications without having to manage the underlying physical hardware. Infrastructure as a Service (IaaS) completes the trio by offering virtualised computing power, servers, and storage on demand (Ahmad et al., 2025:3; Mkhize et al., 2025:5). These services are deployed through public, private, or hybrid cloud environments, each carrying different trade-offs between cost, control, and data security (Jayeola et al., 2022:230).

Small and Medium Enterprises (SMEs) serve as the backbone of the South African economy. Both formal and informal South African SMEs account for more than 90 percent of all businesses, employing approximately 60 percent of the workforce and contributing around 40 percent of Gross Domestic Product (Ngo Ndjama & Van der Westhuizen, 2024:606). The National Development Plan (NDP 2030) further projects that SMEs should create 90 percent of the eleven million jobs the government seeks to achieve by 2030 (South Africa. Department of Small Business Development [DSBD], 2023:12). Given their pivotal economic role, the need for SMEs to operate more effectively and competitively has never been more urgent. The adoption of cloud computing can provide SMEs with the platform for scalability, enabling them to access enterprise-grade tools, reduce operational overhead, and respond to market changes without the constraints of traditional IT infrastructure (Ilahi et al., 2024:57; Mkhize et al., 2025:6).

Despite the well-documented potential of cloud computing to reduce costs and improve operational efficiency, South African SMEs continue to adopt cloud-based services at a pace that is far below their potential (Mkhize et al., 2025:8). Research conducted across the African continent consistently reveals that SMEs face a complex combination of technological, organisational, and environmental barriers that hinder adoption (Jayeola et al., 2022:232; IJSRA, 2024:1445). These barriers are not uniform: an SME in Johannesburg may struggle with high mobile data costs and POPIA compliance concerns, while a business in a rural province may additionally contend with persistent load shedding and a complete absence of digital skills support (Mokwena & Sithole, 2024:14).

Understanding why adoption remains low requires a structured theoretical lens. This study employs the Technology-Organisation-Environment (TOE) framework, originally developed by Tornatzky and Fleischer (1990) and widely used in information systems research, as a lens to identify and categorise the factors that influence cloud computing adoption decisions among South African SMEs (Adane, 2021:4; Santos et al., 2023:Article 2220190). The TOE framework organises adoption influences into three interconnected contexts: technological readiness, organisational characteristics, and the broader environmental landscape, making it particularly well-suited to the multi-dimensional nature of SME cloud adoption barriers in developing economies (Abu Bakar et al., 2024:556; Ilahi et al., 2024:58). This study therefore seeks to propose a framework that can guide South African SMEs towards informed and sustainable cloud computing adoption, contributing both to academic knowledge and to practical policy and business support interventions.

2. Background to the Problem, Aim, Objectives and Research Questions

2.1 Background to the Research Problem

South Africa has one of the most advanced ICT infrastructures on the African continent, yet a significant gap exists between how large corporations leverage technology and how small businesses operate. While major enterprises have the resources to adopt cloud computing in full, most SMEs still function with basic digital tools, limited connectivity, and little knowledge of what cloud services can offer them (IJSRA, 2024:1446). This digital divide reflects deep structural inequalities in access to capital, skills, and infrastructure that have persisted across the post-apartheid period (FinMark Trust, 2024:3).

SMEs are formally defined under the South African National Small Business Act (Act 102 of 1996, as amended) as separately owned and managed businesses that fall within specific thresholds based on employee count and annual turnover (DSBD, 2023:14). In most sectors, a micro-enterprise employs fewer than five people, a small enterprise between six and fifty, and a medium enterprise between fifty-one and two hundred employees. It is within this diverse and economically vital sector that cloud adoption remains critically underdeveloped, creating a significant opportunity for research and intervention.

The deployment models available to SMEs carry real consequences for cost, security, and control. Public cloud services shared across many organisations are the most affordable but offer the least control, while private clouds provide greater security at significantly higher cost. Hybrid models combining both are considered the most practical compromise for most South African SMEs, yet awareness of these options remains low (McKinsey & Company, 2024:2; Ahmad et al., 2025:4). Compounding these technical considerations is the Protection of Personal Information Act (POPIA), which came into full effect in July 2021 and requires that adequate safeguards be in place when personal data is transferred outside South Africa's borders, raising legitimate compliance concerns for SME owners without legal or technical expertise (Mokwena & Sithole, 2024:16).

The TOE framework, first proposed by Tornatzky and Fleischer (1990), provides the most appropriate theoretical lens for this study. Its three interconnected contexts: technological, organisational, and environmental, map directly onto the categories of barriers identified in the literature and have been validated in similar SME cloud adoption studies globally (Jayeola et al., 2022:235; Santos et al., 2023:Article 2220190).

2.2 Research Problem

South African SMEs are not adopting cloud computing at a rate commensurate with its potential to transform their operations, despite the widespread availability of affordable cloud services globally. A study conducted in Polokwane Municipality in 2024 surveyed 230 small businesses and found that the two most prominent barriers were cost and a lack of knowledge about cloud computing services (Mokwena & Sithole, 2024:18).

When SMEs are unable to adopt modern technology, they cannot compete effectively in an increasingly digital marketplace. An SME relying on manual processes and spreadsheets cannot match the efficiency or customer experience offered by a competitor using cloud-based systems. South Africa's unemployment rate exceeded 33 percent in 2024 (Stats SA, 2024), and SMEs remain the largest employer in both the formal and informal economy. The NDP 2030 target of SMEs contributing 60 to 80 percent of GDP cannot be achieved if small businesses remain digitally excluded (DSBD, 2023:15).

If the barriers to cloud adoption are not identified and addressed through a structured framework, South African SMEs will continue to operate below their potential, lose competitive ground to larger and better-resourced businesses, and fail to contribute meaningfully to national economic growth and employment targets. The combined technological, organisational, and environmental factors preventing adoption have not been comprehensively examined through a unified theoretical framework in the South African context, leaving a gap that this study seeks to fill (Adane, 2021:5; Jayeola et al., 2022:240).

Research Problem Statement: *South African SMEs are not adopting cloud computing at a rate that matches its potential to improve their operations, and the combined technological, organisational, and environmental factors preventing adoption in South Africa have not been sufficiently examined through a unified theoretical framework.*

2.3 Aim of the Study

The primary aim of this study is to investigate the technological, organisational, and environmental factors that influence cloud computing adoption among small and medium enterprises in South Africa, using the TOE framework as a theoretical lens, with a view to proposing a framework that can guide SMEs, government departments, and business support organisations in accelerating the transition to cloud-based operations.

2.4 Objectives of the Study

In pursuit of the above aim, this study is guided by the following three objectives:

1. To identify the technological factors that hinder or enable cloud computing adoption among South African SMEs.
2. To examine the organisational and environmental factors that influence cloud computing adoption decisions among South African SMEs.
3. To propose a framework that can guide South African SMEs in making informed and sustainable cloud computing adoption decisions aligned with their specific technological, organisational, and environmental contexts.

2.5 Main Research Question

What are the key technological, organisational, and environmental factors that influence cloud computing adoption among small and medium enterprises in South Africa, and how can these factors be addressed through a contextualised adoption framework?

2.6 Sub-Questions

To operationalise the main research question, the study is further guided by the following sub-questions, each aligned with a specific construct of the TOE framework:

1. What are the technological factors that hinder or enable cloud computing adoption among South African SMEs?
2. What organisational factors influence the cloud computing adoption decisions of South African SMEs?
3. What environmental factors shape SME decisions to adopt or avoid cloud-based systems in South Africa?
4. How can a framework, informed by the TOE lens, be structured to guide South African SMEs in adopting cloud computing in a manner that is affordable, legally compliant, and operationally appropriate for their specific context?

3. Literature Review

3.1 Introduction

This literature review critically evaluates existing knowledge on cloud computing adoption among South African SMEs. Rather than summarising sources descriptively, the review adopts a critical synthesis approach to identify patterns, contradictions, and gaps. It is organised around the Technology-Organisation-Environment (TOE) framework (Tornatzky & Fleischer, 1990), which provides a robust lens for examining multi-dimensional adoption barriers in resource-constrained settings.

3.2 Cloud Computing: Conceptual Background

Cloud computing enables organisations to access computing resources through internet-based services without maintaining extensive physical infrastructure. Through service models such as SaaS, PaaS and IaaS, cloud computing offers SMEs opportunities to improve scalability, reduce costs and enhance operational flexibility. Despite these benefits, adoption among South African SMEs remains limited due to technological, organisational and environmental challenges.

3.3 The Technology-Organisation-Environment (TOE) Framework

Tornatzky and Fleischer (1990:152) propose that enterprise technology adoption is shaped by three interacting contexts. The technological context includes internal and external technologies, their compatibility and complexity. The organisational context encompasses firm size, human capital, and financial resources. The environmental context covers industry competition, government regulation, and infrastructure stability (Abu Bakar et al., 2024:556).

3.4 Technological Factors Influencing Cloud Adoption

Relative advantage refers to the degree to which cloud computing is perceived as superior to existing legacy infrastructure. Ilahi et al. (2024:57) demonstrate that when SME managers recognise cloud configurations lower operational costs and improve collaboration, adoption intent rises. However, Mokwena and Sithole (2024:20) present a contradictory finding from their empirical study in Polokwane Municipality: many localised SME owners do not see a clear relative advantage in cloud migration because they lack basic knowledge about cloud capabilities, viewing it as an unnecessary expense rather than an efficiency driver. This contradiction suggests that perceived advantage is not inherent to cloud technology but depends critically on prior digital literacy.

Compatibility evaluates how well cloud solutions align with existing business processes, workflows, and legacy architectures. Yaseen et al. (2023:320) establish that technological compatibility is the strongest positive predictor of long-term adoption intent among small business owners. Conversely, Mkhize et al. (2025:12) point out that severe incompatibility arises when SMEs rely on specialised on-premises legacy systems that cannot connect to cloud environments, forcing complex modifications that stall migration projects. Complexity represents the perceived difficulty of configuring cloud architectures. While simple SaaS solutions present minimal complexity, IaaS and PaaS demand advanced technical skills. Yaseen et al. (2023:320) emphasise that if IaaS and PaaS configurations are perceived as too complex, this acts as a direct inhibitor, an issue amplified for South African SMEs operating without formal IT support.

Security concerns focus on risks related to data breaches and loss of control. In global studies, security is often framed as a manageable technical challenge. However, Jayeola et al. (2022:228) show that within smaller enterprises, security is frequently viewed as an absolute inhibitor, stemming from distrust of storing sensitive data on shared remote servers. Data privacy and POPIA compliance represent a distinct technological barrier unique to South Africa. POPIA, which came into full effect in July 2021, mandates strict protections for personal data, including restrictions on cross-border transfers. Mokwena and Sithole (2024:22) identify POPIA compliance as a technology-context barrier not replicated at equivalent levels in comparable developing-economy studies from Jordan or Malaysia. Because many global cloud hyperscalers use international data centres, SME owners fear compliance violations. Lacking internal legal expertise, many South African small businesses choose to avoid cloud migration entirely to prevent regulatory penalties.

3.5 Organisational Factors Influencing Cloud Adoption

Top management support refers to the degree to which owners understand and fund technology innovations. Within the SME sector, where organisational structures are often flat and centralised, the personal attitude of the business owner is critical. Abu Bakar et al. (2024:556) note that when an owner values digital transformation, they can overcome internal resistance and align resources. Conversely, if leadership is risk-averse or lacks digital literacy, the enterprise typically remains frozen in legacy models.

Organisational readiness and financial resources examine whether a company has the capital needed for cloud adoption. A common debate concerns the true cost of migration. Proponents argue cloud computing lowers expenses by converting capital investments to operating costs. However, McKinsey and Company (2024:4) counter that capital constraints prevent South African SMEs from absorbing migration costs, even where cloud represents long-term savings. SMEs must pay for internet upgrades, data migration pipelines, dual-system running costs, and consultant fees. For resource-constrained small businesses, these initial costs create a significant financial barrier.

Digital skills and IT expertise represent the foremost organisational barrier in sub-Saharan Africa. Mkhize et al. (2025:14) establish through systematic review that inadequate digital skills and the absence of dedicated IT personnel are critical barriers. Unlike large enterprises with internal IT departments, typical South African SMEs rely on self-taught owners with limited technical training. This lack of internal expertise leaves businesses poorly equipped to evaluate cloud providers, handle data migrations, or troubleshoot network issues, consequently increasing reliance on expensive external consultants and raising total cost of ownership.

3.6 Environmental Factors Influencing Cloud Adoption

Competitive pressure represents the influence felt from digitally-enabled rivals. Jayeola et al. (2022:233) demonstrate that competitive pressure is a primary adoption motivator in SME-dense markets. However, this pressure varies by location: while urban SMEs face intense competition from digital platforms, businesses in remote provinces operate in less tech-dense local markets, reducing perceived urgency to adopt cloud solutions.

Government and vendor support shape the institutional environment. In South Africa, a clear gap exists between policy intentions and actual business support. While the DSBD has drafted policies targeting SME modernisation, ground-level support remains limited. Furthermore, Jayeola et al. (2022:237) and Mokwena and Sithole (2024:25) note that limited local vendor support for SMEs leaves them with minimal technical guidance, forcing small businesses to navigate complex compliance and migration challenges entirely on their own.

Infrastructure constraints: load shedding and connectivity expose a critical difference between global and South African research. International studies treat utility infrastructure as given, but Adane (2021:6) and Mokwena and Sithole (2024:28) emphasise that persistent electricity supply disruptions undermine the reliable connectivity cloud services require. When power cuts take down cellular towers and business routers, cloud-hosted applications become completely inaccessible. This dynamic creates operational vulnerabilities, forcing business owners to maintain expensive on-premises solutions, thereby neutralising the cost benefits of cloud models.

3.7 Critical Synthesis of Literature

Author(s) & Year	Country	Methodology	Main Findings	TOE Dimension(s)
Santos et al. (2023)	Portugal	Quantitative survey	Relative advantage and compatibility strongly predict adoption; cloud integration improves MSME operational performance	Technological, Organisational
Abu Bakar et al. (2024)	Malaysia	Quantitative survey	Competitive pressure and top management support are most influential drivers of cloud adoption intent	Organisational, Environmental
Yaseen et al. (2023)	Jordan	Quantitative survey	Technological compatibility is primary driver; high system complexity acts as dominant inhibitor	Technological
Mokwena & Sithole (2024)	South Africa	Quantitative survey	High costs, poor technical knowledge, and POPIA data privacy concerns act as severe adoption barriers	Technological, Environmental
Mkhize et al. (2025)	Sub-Saharan Africa	Systematic review	Inadequate digital skills, shortage of IT personnel, and high data costs represent most critical barriers	Organisational

Table 1: Comparative Summary of Key TOE-Based Cloud Adoption Studies

3.8 Research Gap

Existing TOE-based studies predominantly target stable developing economies such as Portugal, Malaysia, and Jordan, where continuous electricity supply and mature legal compliance systems are assumed, conditions that do not reflect the South African context. South African scholarship remains fragmented, examining load shedding, POPIA, and digital skills deficits in isolation rather than through an integrated theoretical lens (Adane, 2021:7; Mokwena & Sithole, 2024:30). No contextualised adoption framework tailored to South Africa's specific regulatory, infrastructural, and economic conditions currently exists. This study addresses both gaps by applying a complete TOE lens to propose a practical, context-sensitive adoption framework for South African SMEs.

4. Design, Methodology and Ethics

4.1 Research Design

This study employs a Systematic Literature Review (SLR) design following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines to ensure completeness and transparency in the search, screening, and reporting processes (Page et al., 2021:1). An SLR is the optimal design for resolving multi-dimensional adoption barriers by consolidating fragmented research into an integrated framework. The study evaluates technology implementation as a sociotechnical process embedded within local organisational limits and shifting institutional realities, including electricity disruptions and POPIA constraints.

4.2 Research Methods and Data Sources

To capture a comprehensive range of relevant studies, five electronic databases will be searched: Scopus, Google Scholar, ACM Digital Library, Wiley Online Library, and MDPI. The primary search string uses Boolean operators to ensure alignment with the study's objectives: ("cloud computing adoption" OR "cloud adoption") AND ("SMEs" OR "small and medium enterprises" OR "SMMEs") AND ("Technology - Organisation - Environment" OR "TOE framework") AND ("South Africa" OR "developing economy").

4.3 Study Selection and Inclusion/Exclusion Criteria

Variable	Inclusion Criteria	Exclusion Criteria
Publication window	2019–2026	Pre-2019 studies
Language	English-language publications only	All non-English publications
Document type	Peer-reviewed articles, conference proceedings, and master's/doctoral theses from accredited universities	Editorials, blogs, non-peer-reviewed opinion pieces
Thematic focus	Studies evaluating cloud adoption via the TOE framework	Studies on unrelated sectors or disconnected information systems topics
Geographic scope	South Africa or comparable developing economies	High-income Western market studies (e.g., USA)

Table 2. Study inclusion and exclusion criteria.

4.4 Quality Appraisal

Each study proceeding to full-text review will be appraised using the Mixed Methods Appraisal Tool (MMAT), which accommodates qualitative, quantitative, and mixed-methods designs within a single standardised framework (Hong et al., 2018:7).

4.5 Data Extraction and Synthesis

Extraction parameters include adoption barrier type (mapped to a specific TOE dimension), study country, SME size, research methodology, primary findings, and contextual relevance to the South African regulatory and infrastructure environment.

4.6 Ethics

As a secondary study using published sources, no human participant risks exist. The protocol will be submitted to the CPUT Departmental Ethics Committee for exemption, and official CPUT Harvard referencing guidelines will be maintained throughout.

5. Delineation, Timeline

The study focuses exclusively on South African SMEs employing 200 or fewer individuals in compliance with the National Small Business Act, the three principal cloud service models (SaaS, PaaS, IaaS), three deployment environments, and core TOE constructs only. Research milestones progress systematically from the proposal submission in June 2026, through screening and quality appraisal in August 2026, factor extraction and framework drafting in September 2026, to final synthesis and submission in October 2026.

6. Significance

This study makes two interconnected contributions. Theoretically, it extends TOE-based cloud adoption research into the South African SME context, where infrastructure instability, POPIA compliance obligations, and acute digital skills deficits represent conditions not accounted for in existing global models. The proposed framework addresses this contextual gap by integrating all three TOE dimensions into a unified, locally-relevant adoption guide. Practically, the framework provides SME owners, government departments such as the DSBD, and technology vendors with an evidence-based tool to diagnose adoption barriers and design targeted interventions, directly supporting the NDP 2030 goal of SME-led economic growth and employment creation.

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